2-1-1: Examining the Principles of Security

At the end of this episode, I will be able to:

1. Identify and explain fundamental security concepts, given a scenario.

Learner Objective: *Identify and explain fundamental security concepts, given a scenario*

Description: In this episode, the learner will examine the security triangle and the goals of security. We will also explore methods to accomplish the principles of security.

- What is the security triangle?
  - A diagrammatic was to view the main three goals of security.
- What are the main goals of security?
  - Confidentiality
    - Ensuring that sensitive data is accessible only to authorized individuals or systems
  - Integrity
    - Ensuring data remains accurate, consistent, and unaltered during storage, transmission, and processing.
  - Availability
    - Ensuring authorized users have timely and uninterrupted access to resources and systems.
- Are there concepts to supporting the CIA triangle?
  - Non-repudiation - by ensuring that individuals cannot deny their actions or transactions, enhancing Accountability and contributing to the authenticity and Integrity of information exchanges.
  - authenticity - by confirming the legitimacy of users, data sources, and communications, ensuring that information is accurate, trustworthy, and free from unauthorized alterations.
  - Accountability - by establishing a traceable record of actions and activities that hold individuals responsible for their actions, contributing to the overall security and Integrity of data, systems and communications.
  - Principle of least privilege - restricting user and system permissions to the minimum necessary level reduces potential attack surfaces and safeguards Confidentiality, Integrity, and Availability of resources.
- What are some methods or examples to accomplish these goals?
  - Confidentiality
    - Encryption - by converting sensitive data into a scrambled output using complex algorithms to prevent unauthorized access.
    - Access controls - by regulating and restricting entry to sensitive information, allowing only authorized access to data, systems, or resources.
    - Secure communication channels - by employing encryption and authentication mechanisms that safeguard the transmission of sensitive data from unauthorized interception or eavesdropping.
  - Integrity
    - Hashing - Hashing supports Integrity by generating fixed-size unique values from data, enabling verification of data integrity. Any alterations to the original data would result in a different hash value.
    - Digital signatures - by using cryptographic techniques to associate a unique digital identifier with a message or document, allowing recipients to verify both the sender's authenticity and the document's Integrity.
    - Integrity checks - by comparing the current state of data or systems with a trusted reference, detecting any unauthorized changes or modifications that may have occurred.
  - Availability
    - Redundancy - by duplicating critical components or systems, ensuring that if one fails, another can seamlessly take over, minimizing downtime and maintaining continuous access to resources.
    - Load balancing - by distributing incoming network traffic or workloads across systems, preventing systems from becoming overwhelmed.
    - Backups - by creating copies of data and systems, enabling restoration in the event of data loss, system failures, or other disruptions, ensuring continuity of operations.

- Peform data integrity check to support the CIA Triangle
  - Log in to *ACIWINCLIENT01* (names may vary)

- In the Windows *Taskbar* select the *Microsoft Edge* icon
- In the *Microsoft Edge* window using the address bar browse to and download the *Quickhash#GUI* application:

```
https://www.quickhash-gui.org/downloads/
```

- On the Windows *Taskbar* select the *File Explorer* icon and navigate to the following location:

```
C:\Users\currentuser\downloads
```

- In the *File Explorer* window, right-click snd extract the *QuickHash-GUI-Windows-v**** file to the current location
- In the *Extract Compressed (Zipped) Folders* under *Select a Destination and Extract Files* ensure that the checkbox next to *Show extracted files when complete* is selected and choose *Extract*
- In the *File Explorer* popup window select the *64-Bit* folder and choose the *Quickhash-GUI* application
  - If a *Defender* popup appears select the *More info* hyperlink and choose *Run anyway*
- Minimize the *QuickHash* window
- On the Windows *Desktop* right-click and choose *New > Text Document*
- In the *Untitled - Notepad* window type the following:

```
Secret 1
Secret 2
Secret 3
```

- In the *Untitled - Notepad* window from the options menu select *File > Save as*
- In the *Save As* window save the file to the following location:

```
C:\Users\currentuser\Desktop\version1.txt
```

- Close the *Notepad* window
- Restore the *QuickHash* application
- In the *QuickHash* window select the *Files* tab
- In the *QuickHash > File* window under *Single File Hashing* choose *Select File*
- In the *Open existing file* window select the following file:

```
C:\Users\currentuser\Desktop\version1.txt
```

- In the *QuickHash > File* window under the *Select File* button locate and mark down the *hash value*

```
Example: 24B40CEF852E4CDD764275626D23CEE21AFA85D7
```

  - Note - this value will be checked in a later step to establish that integrity have been maintained.
- Minimize the *QuickHash* window
- On the Windows *Desktop* double-click the *version1.txt* file
- In the *version1.txt* file delete the following:

```
Secret 3
```

- In the *version1.txt* window select *File > Save*
- Restore the *QuickHash* window
- (If necessary) In the *QuickHash* window select the *Files* tab
- In the *QuickHash > File* window under *Single File Hashing* choose *Select File*
- In the *Open existing file* window select the following file:

```
C:\Users\currentuser\Desktop\version1.txt
```

- In the *QuickHash > File* window under the *Select File* button locate and mark down the *hash value*

```
Example: 753A216B99727B425C55CBC15B377C34BA1E0139
```

- In the *QuickHash > File* window under *Expected Hash Value* past the value copied from the previous step.
- In the *Quickhash-GUI* window examine the warning
  - Note - use this value to compare the value marked down in previous step. Notice how the values are different meaning integrity has not been maintained.

- Additional Resources:
  - If applicable